01

PolisProof

AI can summarize a decision.
PolisProof proves what was actually decided.

Evidence-first verification for multilingual public records. Apertus extracts atomic claims; deterministic rules bind them to exact sources, surface contradictions, and block unsupported conclusions.

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APACHE-2.0

THE FAILURE MODE

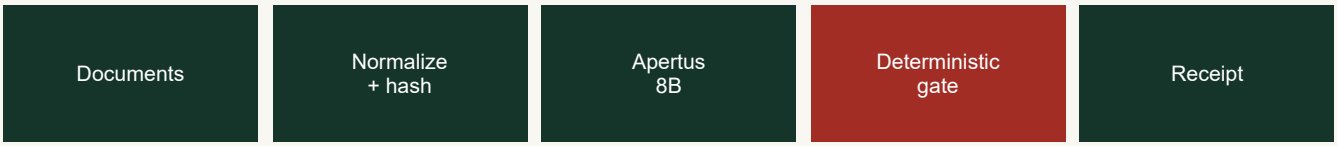
Draft budgets, final minutes and legal notes disagree. Conventional summaries can promote proposals into approvals or silently drop conditions.

THE PROMISE

Every accepted claim points to a verbatim span and source hash. Conflicts stay visible. The application—not the model—owns the verdict.

TRACK FIT AND SOVEREIGN ARCHITECTURE

Designed for control at the deployment boundary



MANDATORY TARGET

PolisProof is deployable on-premise or air-gapped: application code uses the Python standard library and talks to an OpenAI-compatible Apertus server. Open weights keep records and inference inside the operator's boundary.

> **Runtime sovereignty:** local application + locally served Apertus weights + no external network dependency.

DEMONSTRATED INTEGRATION

The hosted Hugging Face router verified compatibility with real Apertus 8B and 70B calls. That hosted path is **not** claimed as sovereign, on-premise, air-gapped, or representative of local performance.

> **Tested limitation:** replay/fallback run offline, but local Apertus weight execution was not tested on this machine.

WHY APERTUS

Apertus provides open weights, multilingual coverage and both 8B and 70B variants. The 8B lane handles routine extraction; only disputed context is eligible for 70B review. This makes the architecture portable, auditable and cost-aware without delegating final authority to an opaque model.

TRUST BOUNDARY

Documents are untrusted data. Prompt-injection markers force review. Model JSON crosses a strict parser boundary. Source-span existence, hashes, amounts, currencies, dates, negation, actors and conditions are checked deterministically.

APERTUS PIPELINE

Selective intelligence, deterministic authority

8B extracts claims | Strict JSON parser | Rules find material dispute | 70B reviews disputed context | Gate emits verdict

Stage	Responsibility	Failure behavior
Apertus 8B	Atomic multilingual extraction with exact source artifact/span.	Malformed or unavailable output fails closed; never fabricated.
Parser	Schema, enum and confidence validation; bounded repair accounting.	`MODEL_OUTPUT_INVALID` remains visible.
Verifier	Evidence coverage and cross-document amount/date/actor/negation/condition checks.	Unsupported high-impact claims are blocked.
Apertus 70B	Review context only for disputed claims.	Unavailable review cannot erase deterministic findings.
Receipt	Hashes, spans, models, routes, tokens, latency, reasons and output hash.	Execution mode preserves provenance.

VERIFIED HOSTED MODEL IDENTIFIERS

```
swiss-ai/Apertus-8B-Instruct-2509:publicai
swiss-ai/Apertus-70B-Instruct-2509:featherless-ai
```

On 12 October, one minimal call to each returned HTTP 200 and `POLISPROOF_OK`: 75 tokens each, 1.747 s (8B) and 7.568 s (70B). Calculated cost was USD 0.000080925. These smoke calls prove execution—not extraction quality.

EXECUTION LABELS

LIVE_APERTUS | RECORDED_REPLAY | OFFLINE_FALLBACK | ERROR

FICTIONAL DEMONSTRATION

€60k proposal ≠ €40k conditional approval

Synthetic evaluation materials. No real person, council, project or allegation is represented.

GERMAN FINAL MINUTES

> *“Der Rat genehmigt … höchstens 40.000 EUR.”*

Approval is capped at EUR 40,000 and subject to legal review.

FRENCH DRAFT ANNEX

> *“Le budget proposé … est de 60 000 EUR.”*

A proposed budget, not the final approval.

ITALIAN FINAL NOTE

> *“L'approvazione è valida solo dopo il controllo legale.”*

Approval is valid only after legal review.

UNSUPPORTED DRAFT CLAIM

> *“Council approved €60,000 unconditionally.”*

Conflicts with both amount and condition evidence.

BLOCKED

AMOUNT_CONFLICT | MISSING_CONDITION

The deterministic result is reproducible from the supplied documents. The bundled replay demonstrates the whole route as `8B → 70B`; its model fixtures are explicitly synthetic and never presented as live inference.

AUDIT OUTPUT

The downloadable JSON receipt includes normalized source hashes, extracted claims, exact evidence spans, reason codes, route events, execution mode and a 64-character SHA-256 output hash.

EVALUATION AND EVIDENCE

What is measured—and what is not

44 / 44

Unit and integration tests passing.

12 / 12

Frozen deterministic rule cases passing.

100%

Contradiction recall on 8 labelled synthetic fault cases.

0%

False-block rate on 1 clean end-to-end replay packet.

Evidence layer	Observed result	Scope
Deterministic corpus	12/12; unsupported block 1/1; evidence coverage 5/5.	Small synthetic set; not population performance.
Routing replay	2/3 packets 8B-only; 1/3 escalated.	Recorded synthetic behavior, not live latency.
Live smoke	Both 8B and 70B returned HTTP 200.	Trivial exact-text prompt only.
Stratified live attempt	6 packets × 3 modes attempted; downstream providers returned Cloudflare 1010 / HTTP 403.	No successful inference; no quality percentages or token charges claimed.

FROZEN LIVE METHODOLOGY

The 30-packet corpus spans English, German, French and Italian across clean, contradiction, condition, unsupported, adversarial and ambiguous categories. The harness preserves raw sanitized output, tokens, latency, schema validity, citations, routing and verdicts.

> **Honest result:** the full live corpus remains incomplete. Extraction precision/recall, structured-output reliability, adversarial success rate and live routing savings are unavailable.

SECURITY, REPRODUCIBILITY AND RELEASE

Submission-ready with explicit owner gates

SECURITY

- Credentials read only from environment.
- No secrets in receipts, fixtures or evidence.
- One-megabyte upload limit and path containment.
- Untrusted documents cannot override the system prompt.
- Failures remain visible and fail closed.

REPRODUCE

```
cd track_2b
make test
make eval
make run
```

Default `make run` is a network-free recorded demo. Live mode uses the verified HF router configuration or a local OpenAI-compatible Apertus endpoint.

LICENSING

- Code: Apache License 2.0.
- Documentation/design: CC-BY-4.0.
- Synthetic data: CDLA-Permissive-2.0.

KNOWN LIMITATIONS

- No local Apertus-weight execution on this machine.
- Live 30-packet benchmark incomplete.
- Exact-span matching can over-escalate harmless formatting.
- Receipt hash is tamper-evident, not a digital signature.

PROJECT LINKS

Repository URL: **pending owner publication**
Demo video URL: **pending owner upload**
Official submission: `https://hackapertus.ch/online-hack/submissions`

REFERENCES

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[4] Hugging Face Inference Providers pricing, huggingface.co/docs/inference-providers/pricing
[5] Featherless request pricing, featherless.ai/docs/request-pricing-and-credits
[6] Apertus model cards, huggingface.co/swiss-ai